

Simon Dumas

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Game developer with a strong background in gameplay programming and systems design, with a released title on Steam, seeking a game development position.

SKILLS

Languages: C#, C++, HLSL, TypeScript, Python

Engines & Platforms: Unity, Unreal Engine, DirectX 11, Netcode for GameObjects, Steam, PlayStation

Tools: Visual Studio, VS Code, GitHub, Trello, Blender, Autodesk Maya, Photoshop

EXPERIENCE

Rochester Institute of Technology – Research Developer

January 2025 – June 2025

- Worked on a GPU accelerated fluid simulation, focusing on data visualization for a grant presentation.
- Researched and presented emerging technologies such as digital twins and Nvidia Omniverse for future implementation.

PROJECTS

Screen Space – Unity, C#

Released on Steam, November 2025

- Designed and developed a territory-based couch co-op shooter in which player health is tied to the player's controlled space.
- Implemented a dynamic map system using Delaunay Triangulation to generate a graph of triangles, transferred as the pieces of player territory at runtime.
- Built CPU opponents that navigate the territory graph across four difficulty levels, along with five ship classes with unique special abilities.
- Developed online multiplayer with Netcode for GameObjects and Unity Relay, using host-authoritative shot validation and client-side prediction to keep matches fair and responsive.
- Created all of the game's art, music, and sound.

Game Graphics Engine – C++, DirectX 11, HLSL

github.com/SimonDumas79/IGME-540

- Built a 3D graphics engine in DirectX 11 and C++ supporting lighting, physically based rendering, normal mapping, shadow mapping, and post-processing.
- Wrote shaders for PBR, normal mapping, shadows, blur, pixelation, and posterization.

EDUCATION

Rochester Institute of Technology – Bachelor of Science in Game Design and Development

2026

Relevant coursework: Data Structures & Algorithms I & II, Foundations of Game Graphics Programming, Console Development, Game Design & Development I & II